

Vessel Log On — Functional Specification

Version: 0.4 (draft) **Status:** For review **Domain:** Marine rescue vessel log on, watch, and log off **Audience:** Anyone implementing or evaluating a system that performs this function

1. Purpose and scope

1.1 Purpose

This document specifies the required behaviour of a system that records vessels departing under the watch of a marine rescue unit, monitors their safe return, and escalates when they do not return.

It is written to be implementation-independent. It names no product and assumes no technology. It is equally usable as a build specification and as an evaluation checklist for an existing system.

1.2 The primary safety objective

All requirements in this document derive from one objective:

PSO. If a vessel does not return when it said it would, the unit shall know, and the unit shall be able to find it.

Where any two requirements appear to conflict, the one that better serves PSO prevails.

1.3 Scope

In scope: capture of log on details across all channels; identification and verification of vessels and persons; search across unit records; monitoring of active log ons; overdue detection and escalation; log off; amendment; record retention and audit.

Out of scope: conduct of the search or rescue itself; asset and crew management; member administration and billing; incident management beyond the point of escalation.

1.4 Priority

Requirements carry a priority reflecting current operational need, not importance:

Priority	Meaning
P1	Primary focus. Capture and verification of radio log ons — §5, §6, §7.
P2	Required, but adequately served by existing practice — §8, §9.
P3	Required for completeness; lowest urgency.

1.5 Requirement language

Shall — mandatory. **Should** — strongly recommended; deviation requires justification. **May** — optional.

2. Definitions

Term	Definition
Log on	A record that a vessel has departed under the unit's watch, with a stated intention to return by a stated time.
Log off	The act of closing a log on because the vessel has returned or otherwise ended its trip.
Operator	A person, usually a volunteer, receiving log ons at a unit.
Unit	A marine rescue base holding the watch for a geographic area.
Member	A person with a standing record held by the organisation.
Public user	A person without a standing record, logging on as a non-member.
Identifier	Any value that resolves to a stored record: member number, vessel registration, mobile number, vessel name.
Verification	Confirmation that captured identifying data corresponds to a real, correct record.
Cross-verification	Verification achieved by two independent identifiers resolving to the same record. See §7.
Draft	A log on that has been created and persisted but not yet completed.
Active	A log on for a vessel currently out, before its ETA.
Overdue	An active log on whose ETA has passed without log off.
Escalated	An overdue log on for which the escalation sequence has begun.
ETA	The time by which the vessel states it will return or report. Drives overdue detection.
POB	Persons on board.
Enrichment	Addition of detail to an existing log on after its initial creation.
Airtime	Occupancy of a shared radio channel. A finite, contended resource.

3. Domain model

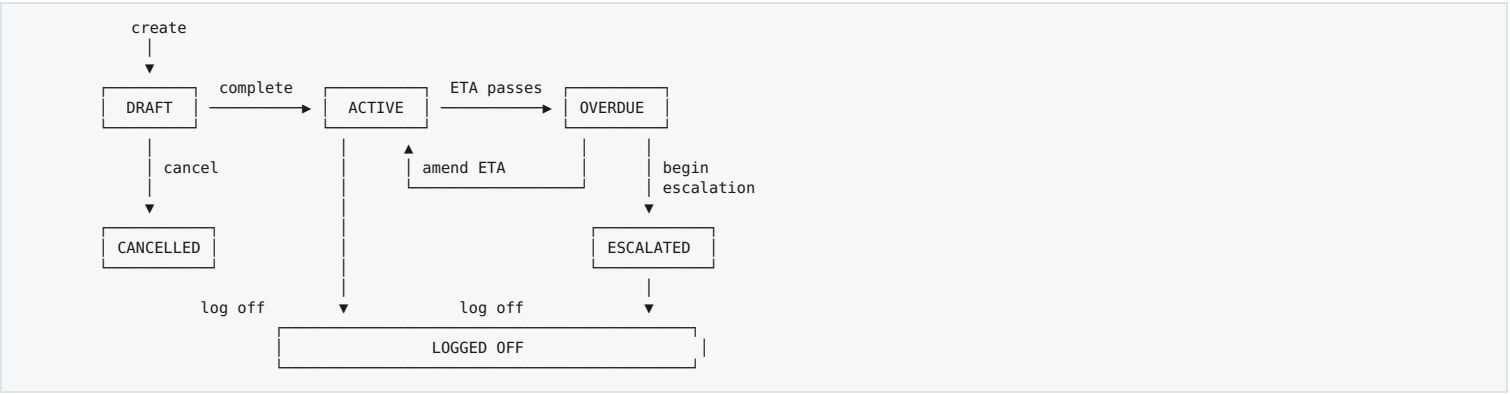
3.1 Entities

Entity	Description	Key attributes
LogOn	The central record. One per trip.	state, channel, operator, unit, departure point, departure time, destination, ETA, POB, verification state, created/updated timestamps
Vessel	A boat known to the system.	registration, name, length, hull colour, type (power/sail/other), make, model, AIS identifier
Member	A person with a standing record.	member number, name, contact numbers, associated vessels
PublicUser	A person without a standing record.	name, contact numbers, associated vessels
OnboardContact	A person aboard or ashore who can be reached.	name, relationship, contact number
Identifier	A captured identifying value and its resolution outcome.	type, value as captured, resolved record, verification state
EscalationStep	One action taken against an overdue log on.	action, timestamp, operator, outcome
Location	A departure or destination point.	name, coordinates (optional), free-text flag
Operator	The person taking the call.	identity, unit, shift

3.2 Relationships

- A **LogOn** references at most one **Vessel** and at most one **Member** or **PublicUser**. All three may be absent in a Draft.
- A **LogOn** holds zero or more **Identifier** records — one per identifying value captured, retaining the value **as captured** alongside its resolution.
- A **LogOn** holds zero or more **OnboardContact** records.
- A **LogOn** holds zero or more **EscalationStep** records, ordered by time.
- A **Vessel** may be associated with zero or more **Members** and **PublicUsers**, and vice versa.

3.3 States and transitions



Transition	Trigger	Notes
→ Draft	Operator begins capture	Persisted immediately (CAP-1)
Draft → Active	Operator marks complete, or ETA and POB present and operator confirms	Draft may remain Draft indefinitely
Draft → Cancelled	Vessel did not depart	
Active → Overdue	ETA passes without log off	Automatic, system-driven
Active → Active	Amendment of ETA, destination, POB	Appends, does not overwrite (REC-3)
Overdue → Active	Vessel makes contact, new ETA agreed	
Overdue → Escalated	Escalation sequence begun	
Any → Logged off	Vessel returns or trip ends	Explicit operator action only (WAT-7)
Active → Transferred	Another unit assumes the watch	Record moves without re-keying

A **Draft** is a valid, legally-held, searchable record in every state-dependent behaviour except overdue detection, which requires an ETA.

3.4 Log on variants

Variant	Difference
Standard	Single departure, single ETA.
Long term	Multi-day passage. ETA replaced by a schedule of expected position reports; overdue triggers on a missed report.
Bar crossing	Short, high-risk watch over a specific hazard. Short ETA. May be nested within a standard log on.

All variants are the same entity with different ETA semantics. They shall not be modelled as separate record types.

4. Operating context

Context that constrains the requirements. Stated as fact, not as complaint.

OC-1. Log ons arrive by marine radio, telephone, in person, and self-service. Radio is the dominant channel and the most constrained; the system shall be designed for radio and will thereby satisfy the others.

OC-2. A radio log on is an interview, not a dictation. The caller states a rough intention; the operator elicits the remaining detail by asking. Field order is arbitrary and varies per call.

OC-3. There is no published procedure instructing callers what to say, and no enforced script instructing operators what to ask. Operators are volunteers of varying experience and each works differently. The system shall not assume procedural consistency.

OC-4. Airtime is contended. Every question the operator asks occupies a shared channel that other vessels are waiting on. The system shall treat operator questions as a cost to be minimised.

OC-5. Phonetic alphabet use is inconsistent, by callers and operators alike. Spoken digits are reliable, because the operator controls the conversation and can request a repeat. **Spoken letters are not**, and confuse along the predictable set in Appendix B.

OC-6. Identifying detail — vessel registration, mobile number — typically arrives late in a call or only when asked, and sometimes on a later call. It shall not be required early.

OC-7. The organisation is under a legislative obligation to hold a computer record of log ons. A record held only on paper does not discharge that obligation.

OC-8. Calls arrive in bursts. Peak load is a period of good weather, which is also when the greatest number of vessels are at sea.

5. Requirements — Capture (P1)

CAP-1. The system **shall** persist a log on record containing any subset of fields, including an empty subset, from the moment capture begins.

Rationale: a record that exists partially is operationally and legally superior to one that exists only on paper (OC-7). Blocking creation converts a survivable gap into a missing record.

CAP-2. The system **shall not** require any field to be populated in order to create, save, or retain a log on. *Rationale: any mandatory field is a point at which capture can fail entirely (PSO).*

CAP-3. The system **shall** accept field entry in any order, with every field reachable and editable from the keyboard without pointing-device interaction. *Rationale: OC-2. Entry order is dictated by the caller.*

CAP-4. The system **shall** leave unpopulated fields empty, and **shall not** pre-populate any field with a value that has not been supplied. *Rationale: an unsupplied value that looks supplied produces a record that appears correct and is not — the most dangerous output of the system.*

CAP-5. Where the system detects an inconsistency — for example an ETA preceding the departure time — it **shall** present the warning adjacent to the field concerned, at the time of entry, and **shall** permit the record to be saved regardless. *Rationale: inconsistency is information; preventing the save discards the record to protect the field.*

CAP-6. The system **shall** permit multiple log ons to be in Draft simultaneously, and **shall** permit an operator to suspend and resume capture of any Draft without loss. *Rationale: a priority radio call may interrupt capture at any point.*

CAP-7. The list of active and overdue log ons **shall** remain visible to the operator during capture of a new log on. *Rationale: situational awareness is the operator's primary function; capture must not suspend it.*

CAP-8. The system **shall** accept time input in the forms operators speak and write, including 24-hour (1500), 12-hour (3pm), and relative (+2h). *Rationale: format conversion is cognitive load during a timed task.*

CAP-9. The system **shall** accept free text for departure point and destination, and **shall** offer known locations as suggestions without restricting entry to them. *Rationale: callers name places that exist in local usage and in no list.*

CAP-10. The system **shall** display, during capture, which fields remain unpopulated. *Rationale: OC-3. In the absence of a procedure, the visible gap list is the procedure.*

CAP-11. The system **shall** rank unpopulated fields by the priority in §6.1, giving greatest prominence to fields serving PSO and least to vessel description. *Rationale: OC-4. Undifferentiated prompting spends airtime on low-value fields.*

CAP-12. Prompting under CAP-10 and CAP-11 **shall** be advisory, and **shall not** prevent an operator from completing or closing a capture. *Rationale: the operator, not the system, judges what is worth a transmission.*

CAP-13. The system **shall** permit any field of an existing log on in any open state to be populated or amended, with the same interaction cost as initial capture. *Rationale: OC-6. Late-arriving detail is the normal case.*

CAP-14. Capture of a log on for a person without a standing record **shall** proceed without first classifying the caller, and **shall not** require vessel description fields before the log on is created. *Rationale: classification is a decision the operator cannot make until identity is known, which is late (OC-6).*

CAP-15. Where vessel or contact detail has been captured previously for the same vessel or person, the system **shall** offer it rather than requesting it again. *Rationale: OC-4. Re-asking known detail spends airtime for no information gain.*

5.1 Capture performance

CAP-16. For a caller whose identifiers resolve to an existing record, and who supplies all requested detail without repetition, a complete log on **shall** be capturable in **30 seconds or less**, keyboard only.

CAP-17. For a caller with no existing record, a complete log on **shall** be capturable in **90 seconds or less**.

CAP-18. Capture performance **shall** be measured against real calls under peak load (OC-8), not estimated or measured under simulation.

6. Requirements — Data (P1)

6.1 Field priority classes

Class	Purpose	Fields
A	Locate the vessel	ETA, POB, destination, departure point, departure time
B	Identify and verify	Member number, vessel registration, mobile number
C	Reach the vessel	Radio channel monitored, onboard/shore contact, AIS identifier
D	Describe the vessel	Length, hull colour, type, make, model

DAT-1. Class A fields **shall** be treated as the minimum useful log on. A record lacking an ETA **shall** be visibly distinguished, since it cannot become overdue.

DAT-2. The mobile number **shall** be treated as both a Class B identifier and a Class C contact, and prioritised accordingly. *Rationale: it serves double duty — it identifies the caller and is the first action on overdue.*

DAT-3. Class D fields **shall not** be requested during capture where the vessel resolves to an existing record carrying them.

DAT-4. The system **shall** retain Class D detail captured for a person without a standing record, and associate it with the vessel for reuse.

DAT-5. Every captured identifier **shall** be stored as captured, independently of whatever record it resolves to. *Rationale: the value the operator heard is evidence; overwriting it with the resolved value destroys the ability to detect a mis-resolution later.*

7. Requirements — Identification, verification and search (P1)

7.1 Cross-verification

IDV-1. The system **shall** support the capture of two or more independent identifiers against a single log on. *Rationale: two identifiers resolving to the same record verify each other. This is the unit's existing accuracy practice and the strongest verification available at no airtime cost.*

IDV-2. On capture of a second identifier, the system **shall** resolve both and **shall** display the outcome immediately, as one of:

Outcome	Condition
Verified	All captured identifiers resolve to the same member and vessel
Conflict	Identifiers resolve to different records
Partial	At least one resolves, at least one does not
Unverified	Fewer than two identifiers captured, or none resolve

IDV-3. A **Conflict** outcome **shall** be presented prominently at the moment of detection. *Rationale: a conflict is the only reliable in-call signal that a captured value is wrong.*

IDV-4. The verification outcome **shall** be stored on the log on and **shall** be displayed wherever the log on appears, including the active watch list.

IDV-5. A verification outcome other than **Verified shall not** prevent the log on being created, saved, or activated. *Rationale: an unverified record is still a record of a vessel at sea.*

7.2 Tolerant resolution

IDV-6. Identifier resolution **shall** return near matches in addition to exact matches, matching across the letter confusion set in Appendix B. *Rationale: OC-5. A registration heard once without phonetics is routinely one letter from correct, and the error is predictable.*

IDV-7. Resolution results **shall** include sufficient descriptive detail — vessel name, length, hull colour, type, associated member — to allow the operator to confirm identity by description. *Rationale: confirming "the white centre console, Sea Breeze" is one short transmission and is more reliable than a phonetic readback of the registration.*

IDV-8. Where resolution returns more than one candidate, all candidates **shall** be presented together.

7.3 Search

SRCH-1. The system **shall** provide a single search input that returns results across all record types: members, public users, vessels, active log ons, draft log ons, and historical log ons. *Rationale: cross-verification (IDV-2) requires two identifiers resolvable in one action. Separate searches per record type make verification cost more airtime than it saves, and it is then not performed.*

SRCH-2. Search **shall** accept any identifier type in the same input without the user selecting a type in advance: member number, vessel registration, mobile number, vessel name, person name.

SRCH-3. A vessel or person **shall** remain returnable by search regardless of whether they currently hold an active log on. *Rationale: an in-progress trip is the state in which lookup is most often needed.*

SRCH-4. Search results **shall** indicate the record type and current state of each result, and **shall** indicate whether a returned vessel or person is currently logged on.

SRCH-5. Search **shall** return results on partial input.

SRCH-6. Every search result **shall** offer a single action to apply it to the log on in progress.

8. Requirements — Watch and overdue (P2)

WAT-1. The system **shall** display a persistent list of active and overdue log ons, ordered by proximity to ETA.

WAT-2. The system **shall** indicate log ons approaching their ETA before that ETA passes.

WAT-3. Transition to Overdue **shall** produce an alert that does not depend on the operator observing a change in a list.

WAT-4. The system **shall** present the escalation sequence as an ordered set of actions, and **shall** record each action attempted with time, operator and outcome. *Rationale: OC-3. Where no written procedure exists, the recorded sequence is the procedure and the audit trail simultaneously.*

WAT-5. On transition to Overdue, the system **shall** surface the log on's verification outcome (IDV-2). *Rationale: personnel about to act on the record need to know which parts of it are verified.*

WAT-6. The system **shall** transfer the complete state of all active log ons, including escalation history, on shift handover, without re-keying.

WAT-7. A log on **shall** leave the active list only by explicit operator action, recorded against that operator.

WAT-8. The system **shall** support transfer of an active log on to another unit, preserving the record and its history without re-keying.

9. Requirements — Record and audit (P2)

REC-1. The system **shall** constitute the record of log ons for the purpose of the organisation's legislative obligation (OC-7), including retention, integrity and export.

REC-2. Every log on and every subsequent action **shall** record the operator, unit, timestamp and channel.

REC-3. Amendments **shall** be appended. Prior values **shall** remain retrievable. *Rationale: post-incident review requires knowing what was known at each point in time, not only the final state.*

REC-4. Verification outcomes and the identifiers as captured (DAT-5) **shall** form part of the permanent record.

REC-5. The system **shall** report on operational measures including call volume by period, overdue frequency, escalation outcomes, and **the proportion of log ons captured during the call versus entered afterwards**. *Rationale: the latter is the direct measure of whether capture is keeping pace with the radio.*

10. Acceptance criteria

Each criterion is a pass/fail test against a running system.

10.1 Capture

#	Test	Pass condition
AC-1	Begin a log on, enter nothing, save.	Record persists and is retrievable.
AC-2	Enter only POB and destination. Save. Close the session. Reopen.	Record persists with those two fields.
AC-3	Open a new log on form and inspect every field.	No field holds a value that was not entered.
AC-4	Enter an ETA earlier than the departure time. Save.	Warning shown beside the field; save succeeds.
AC-5	Begin a log on, start a second, return to the first.	Both retained, neither altered.
AC-6	Capture a log on while the active list is on screen.	Active list remains visible throughout.
AC-7	Complete a full log on using keyboard only.	Achievable; no pointing device required.
AC-8	Enter 1500 , 3pm , +2h into an ETA field.	All three accepted.
AC-9	Enter a destination not present in any list.	Accepted as entered.
AC-10	Time a known-member log on under peak load.	≤ 30 seconds (CAP-16).
AC-11	Time an unknown-vessel log on under peak load.	≤ 90 seconds (CAP-17).
AC-12	Add a registration to a log on created an hour earlier.	Same interaction cost as initial capture.

10.2 Verification and search

#	Test	Pass condition
AC-13	Enter a member number and a registration belonging to the same member.	Outcome shown as Verified .
AC-14	Enter a member number and a registration belonging to different members.	Outcome shown as Conflict , prominently.
AC-15	Enter a registration one letter wrong, within the Appendix B confusion set.	Correct vessel returned as a candidate.
AC-16	Inspect any resolution result.	Includes vessel name, length, colour, type, associated member.
AC-17	Search a registration for a vessel currently logged on.	Vessel returned, marked as currently logged on.
AC-18	Search a member number, a registration, a mobile number and a vessel name in the same input.	All return results without changing search mode.
AC-19	Save a log on with a Conflict outcome.	Save succeeds; conflict remains visible on the record.
AC-20	View the active watch list.	Verification outcome visible per row.

10.3 Watch, record and audit

#	Test	Pass condition
AC-21	Allow an ETA to pass.	Overdue alert raised without operator action.
AC-22	Trigger an overdue on an unverified record.	Verification state surfaced with the alert.
AC-23	Amend an ETA, then inspect history.	Both original and amended values retrievable.
AC-24	Perform a shift handover.	All active log ons and escalation history transfer without re-entry.
AC-25	Run the operational report.	Proportion of live-captured versus later-entered log ons is reported.

10.4 System-level acceptance

AC-26. Operators cease maintaining a parallel paper log, without being instructed to.

This is the single sufficient test. A system passing AC-1 to AC-25 and failing AC-26 has met the letter of the specification and not its purpose.

Appendix A — Justification from current practice

Evidence that the requirements above address real conditions rather than hypothetical ones. Recorded from a unit operating a commercial resilience-management platform alongside a handwritten radio log.

A.1 The paper log is the working record. Its columns are: date/time of call; member number or vessel registration; contact mobile; POB; departure point; destination; return date; return time; *entered into the computer system*; *system record ID*; logged off.

The presence of the last-but-two and last-but-one columns is decisive. **The paper log tracks the computer system as an outstanding task.** The operational record is paper; the computer record is a downstream transcription. Given OC-7, the legally required record therefore runs permanently behind the operational one. AC-26 exists to detect when this has been reversed.

A.2 All-or-nothing saving produces the backlog. The current system cannot persist an incomplete log on. An operator who has captured half the detail has nothing to save, so the entry stays on paper until there is time to complete it — and the backlog grows fastest precisely under peak load (OC-8), when it is least affordable. CAP-1 and CAP-2 address this directly.

A.3 Mandatory fields are the fields that arrive last. Registration and mobile number are required before a record can be saved, and are the two fields that typically arrive last in a call or on a later call (OC-6). The form demands first what the conversation surrenders last. CAP-2, CAP-13 and CAP-14 address this.

A.4 Default values that fail validation. The observed new-log-on form pre-populates departure time and ETA with the same value, then rejects the record on the grounds that the ETA must be later than the departure. The form's initial state fails its own validation, so every log on begins from an error condition. Worse, the paired defaults could be saved unnoticed if validation did not object, producing a record that would never become overdue. CAP-4 and CAP-5 address this.

A.5 Validation errors are remote from their fields. Errors are presented as a block at the top of the form, referring to fields several sections away, discovered only on attempting to save. CAP-5 addresses this.

A.6 Capture occludes the watch. New log on capture opens as a full-screen modal, concealing the list of vessels currently at sea for its duration. CAP-7 addresses this.

A.7 Search is partitioned, and cross-verification is thereby defeated. Members, vessels and active log ons are held behind separate searches. A vessel holding an active log on is not returned by vessel search — the state in which lookup is most often needed. Consequently, cross-verifying two identifiers requires several lookups across separate screens, and under load it is not performed. The unit's principal accuracy mechanism therefore stops operating exactly when accuracy matters most. SRCH-1 to SRCH-4 address this.

A.8 Two identifiers are the existing accuracy practice. Operators routinely request two identifying values — member number and registration, or registration and mobile number. This is not redundancy but a checksum: agreement verifies, disagreement detects an error that is otherwise invisible. The practice is undocumented and unsupported by software, and is performed by eye across the partitioned searches described in A.7. IDV-1 to IDV-5 formalise it.

A.9 Phonetic discipline is not reliable under load. Requesting a phonetic repeat costs a transmission on a contended channel. Under peak load, operators reasonably omit it. Registrations captured from a single unphonetic hearing are therefore routine, and IDV-6 exists because discipline is not an available solution.

Appendix B — Letter confusion set

Errors in unphonetic spoken letters are not uniformly distributed. Resolution under IDV-6 shall treat members of each group as candidate substitutions for one another.

Group	Members
Rhyming ("E-set")	B · C · D · E · G · P · T · V · Z · 3
Nasal	M · N
Sibilant	F · S · X · 6
Long-A	A · J · K · 8
Letter/digit collision	0 ↔ O · 1 ↔ l ↔ L · 5 ↔ S · 2 ↔ Z

Spoken digits are excluded: the operator controls the conversation and may request a repeat, and digit confusion is not systematic (OC-5).

Appendix C — Traceability from version 0.3

0.3	0.4
§1 safety contract	§1.2 PSO
§1.2 legal record	OC-7, REC-1
§2.2 interview	OC-2
§2.3 no procedure	OC-3
§2.4 phonetics	OC-5, Appendix B
§3.1 tiers	§6.1 classes A–D
§3.2 checksum	IDV-1 ... IDV-5
§3.3 unified search	SRCH-1 ... SRCH-6
C1, C2	CAP-1, CAP-2
C3	CAP-4
C4	CAP-5
C5	CAP-6
C6	CAP-7
C7	CAP-16 ... CAP-18
C8	CAP-8, CAP-9
C9, C10	CAP-10, CAP-11, CAP-12
C11	IDV-6 ... IDV-8
C12	IDV-1 ... IDV-5
C13	CAP-13
C14	SRCH-1 ... SRCH-6
W1 ... W7	WAT-1 ... WAT-8
R1 ... R5	REC-1 ... REC-5
§7 anti-requirements	Dissolved into positive requirements; evidence moved to Appendix A
§8 acceptance test	AC-26

Appendix D — Open questions

- On a **Conflict** outcome (IDV-2), what resolution does the operator perform today? Determines whether IDV-3 requires a defined workflow or a flag alone.
- Is a logged-on vessel excluded from vessel search by query construction or by data model? Determines the difficulty of SRCH-3.
- What is the typical transcription backlog — count of entries, and elapsed time from call to computer entry? Establishes the baseline for REC-5 and AC-26.
- Which identifier pair is most commonly used: member number with registration, or registration with mobile number? Determines what IDV-2 optimises for.
- What is the legislative instrument underlying OC-7? Determines retention period and integrity obligations under REC-1.
- For long term log ons (§3.4), what reporting schedule is expected, and what constitutes a missed report?